Introduction to Large Language Models
Assignment- 1
TYPE OF QUESTION: MCQ

Number of questions: 10 **Total mark: 10 X 1 = 10**

QUESTION 1:

Consider the word “paper” in the following sentences:
She submitted her research paper.
He wrapped the gift in paper.

The word “paper” is an example of:

a. Homonymy
b. Synonymy
c. Polysemy
d. Antonymy

Correct Answer: c

QUESTION 2:
Based on Distributional Semantics, which of the following statements is/are true?
(i) The meaning of a word is defined by its relationship to other words.
(ii) The meaning of a word does not rely on its surrounding context.
a. Both (i) and (ii) are correct
b. Only (ii) is correct
c. Only (i) is correct
d. Neither (i) nor (ii) is correct
Correct Answer: d
Solution: According to Distributional Semantics, the meanings of words can be derived from their distributional properties in a large corpora of text. It relies on the context in which words appear.

QUESTION 3:
The sentence “Ravi saw the man with the binoculars.” is ambiguous because of:

a. Lexical ambiguity
b. Morphological ambiguity
c. Syntactic ambiguity
d. Pragmatic ambiguity

Correct Answer: c

Solution:
The prepositional phrase “with the binoculars” can attach to either Ravi or the man, hence the type of ambiguity is syntactic.

QUESTION 4:
Consider the sentences:
Anita met Kavya after work. She was tired.

Who does “she” most likely refer to?

a. Anita
b. Kavya
c. Both equally
d. Cannot be determined

Correct Answer: d
Solution: It is difficult to resolve whether the pronoun “she” refers to Anita or Kavya without additional context.

QUESTION 5:
What is the correct morpheme breakdown of the word “unhappiness”?

a. un + happy + ness
b. unhappi + ness
c. un + happi + ness
d. unhappiness

Correct Answer: a

QUESTION 6:
In the sentence:
“The chef cooked pasta for the guests.”

What is the semantic role of “the guests”?

a. Agent
b. Patient
c. Beneficiary
d. Instrument

Correct Answer: c

QUESTION 7:
Below is a sentence with lexical ambiguity.

Rose rose to put rose roes on her rows of roses.

How many senses of rose are present in the sentence?
a. 2
b. 3
c. 1
d. 4

Correct Answer: d
Solution: Rose[1] rose[2] to put rose[3] roes on her rows of roses[4].

The word rose has four senses:

1. Proper Noun: A person named Rose
2. Verb: stood
3. Adjective: pink-colored (describing roes)
4. Noun: Flower

QUESTION 8:
What issues can be observed in the following text?

> On a much-needed #workcation in beautiful Goa. Workin & chillin by d waves!

a. Idioms
b. Non-standard English
c. Tricky Entity Names
d. Neologisms

Correct Answer: b,d
Solution: We can observe usage of non-standard English such as workin and chillin, neologisms such as workcation in the text.

QUESTION 9:
Sentence 1: Arjun bought a brand-new laptop.
Sentence 2: Arjun owns a laptop.

a. Entailed
b. Contradicted
c. Neutral

Correct Answer: a

QUESTION 10:

Which relation holds between “engine” and “car”?

a. Synonymy
b. Antonymy
c. Meronymy
d. Hyponymy

Correct Answer: c
Solution: An engine is a part of a car.

Introduction to Large Language Models
Assignment- 2
TYPE OF QUESTION: MCQ

Number of questions: 6 **Total mark: 3 X 1 + 2 X 2 + 1 X 3 = 10**

QUESTION 1: [1 mark]
A language model that conditions each word on the previous three words is equivalent to which Markov model?

a. First-order
b. Second-order
c. Third-order
d. Fourth-order

Correct Answer: c
Solution: Please refer to lecture slides.

QUESTION 2: [1 mark]
Which smoothing technique leverages the number of unique contexts a word appears in?
a. Good-Turing
b. Add-k
c. Kneser-Ney
d. Absolute Discounting
Correct Answer: c
Solution: Kneser-Ney uses continuation probability which counts the number of unique left contexts.

QUESTION 3: [1 mark]
Using a language model inside a machine translation system to measure BLEU score is an example of:

a. Intrinsic evaluation
b. Extrinsic evaluation
c. Perplexity evaluation
d. Likelihood estimation

Correct Answer: b
Solution: Performance is measured on a downstream task, hence extrinsic evaluation.

For Question 4 to 6, consider the following corpus:
<s> the wizard sees the dragon </s>
<s> the dragon breathes fire </s>
<s> the wizard casts a spell </s>

QUESTION 4: [2 marks]
Assuming a bi-gram language model, calculate the probability of the sentence:

<s> the dragon casts fire </s>
Ignore the unigram probability of P(<s>) in your calculation.
a. 2/37
b. 1/27
c. 0
d. 1/36
Correct Answer: c
Solution:

P(<s> the dragon casts fire </s>) =

P(the | <s>) x P(dragon | the) x P(casts | dragon) x P(fire | casts) x P(</s> | fire)

P(casts | dragon) = P(fire | casts) = 0

P(<s> the dragon casts fire </s>) = 0

QUESTION 5: [3 marks]
Using add-one smoothing, compute the probability of the sentence:
<s> the dragon casts fire </s>
Assume vocabulary size V includes all unique words (excluding <s>, </s>).
a. 1/7150
b. 1/35
c. 1/67
d. 1/9900
Correct Answer: a
Solution:
P(<s> the dragon casts fire </s>) =

P(the | <s>) x P(dragon | the) x P(casts | dragon) x P(fire | casts) x P(</s> | fire)

Unique tokens:
the, wizard, sees, dragon, breathes, fire, casts, a, spell

<s> the wizard sees the dragon </s>
<s> the dragon breathes fire </s>
<s> the wizard casts a spell </s>

Vocabulary size V= 9

$$P(w_i \mid w_{i-1}) = \frac{Count(w_{i-1}, w_i) + 1}{Count(w_{i-1}) + V}$$

P(the | <s>) = 3+1/3+9 = 4/12 = 1/3
P(dragon | the) = 2+1/4+9 = 3/13
P(casts | dragon) = 0+1/2+9 = 1/11
P(fire | casts) = 0+1/1+9 = 1/10
P(</s> | fire) = 1+1/1+9 =2/10 = 1/5

P(<s> the dragon casts fire </s>) =1/3 x 3/13 x 1/11 x 1/10 x 1/5 = 1/7150

QUESTION 6: [2 marks]
Using the smoothed probability, compute the perplexity of:
<s> the dragon casts fire </s>
(Exclude <s> and </s> from word count.)

a. $9000^{1/5}$
b. $7150^{1/4}$
c. $1/35^{1/5}$
d. $1/9900^{1/4}$
Correct Answer: b
Solution:
Sentence length N=4
Probability P=1/7150
Perplexity=$(1/7150)^{-1/4} = 7150^{1/4}$

Introduction to Large Language Models
Assignment- 3
TYPE OF QUESTION: MCQ

Number of questions: 10 **Total mark: 10 X 1 = 10**

Question 1:

Which component of a neural network introduces non-linearity into the model?

a. Weights
b. Bias
c. Activation function
d. Loss function

Correct Answer: c
Solution: Activation functions introduce non-linearity, enabling neural networks to learn complex patterns beyond linear relationships.

Question 2:

If all activation functions in an MLP are linear, what will be the effect on the network?

a. It becomes equivalent to logistic regression
b. It becomes equivalent to a single linear transformation
c. It will not converge
d. It will overfit

Correct Answer: b
Solution: A composition of linear functions is still linear, so multiple layers collapse into one linear transformation.

Question 3:

Which activation function is least likely to suffer from vanishing gradients?

a. Tanh
b. Sigmoid
c. ReLU

Correct Answer: c

Solution: Its gradient is 1 for positive input and 0 for negative input, so it allows gradients to flow effectively

Question 4:

What happens to dropout during inference?

a. Dropout is applied with higher probability
b. Dropout is applied with lower probability
c. Dropout is disabled
d. Dropout is replaced by L2 regularization

Correct Answer: c
Solution: Dropout is only used during training

Question 5:

Why can a single-layer perceptron not solve the XOR problem?

a. XOR requires probabilistic outputs
b. XOR is non-linear and not linearly separable
c. XOR has more than two classes
d. XOR requires backpropagation

Correct Answer: b
Solution: XOR cannot be separated by a single straight line, so it requires at least one hidden layer.

Question 6:
Among the following options, select the statements that are TRUE about backpropagation.

a. It updates weights using only forward pass
b. It is used to randomly update the weights of a neural network
c. It applies the chain rule from output to input layers
d. It works only for sigmoid activations

Correct Answer: c
Solution: Backpropagation efficiently computes gradients by applying the chain rule backward through the network.

Question 7:
Which type of regularization encourages sparsity in the weights?
a. L1 regularization
b. L2 regularization
c. Dropout
d. Early stopping
Correct Answer: a
Solution: L1 regularization encourages sparsity in the weights.

Question 8:
What can one observe during the training of a neural network if the learning rate in gradient descent is set too high?

a. Loss decreases smoothly
b. Loss oscillates or diverges
c. The loss always stays zero
d. None of these

Correct Answer: b

Question 9:

State the reason for softmax being commonly used in multi-class classification.

a. Outputs values in $[-1,1]$
b. Produces probabilities that sum to 1
c. Avoids overfitting
d. Eliminates the need for regularization

Correct Answer: b

Question 10:

What is the primary goal of regularization in machine learning?
a. To speed up the training process and help in faster convergence
b. To reduce overfitting and improve generalization
c. To help increase the model complexity
d. None of these

Correct Answer: b

Introduction to Large Language Models
Assignment- 4
TYPE OF QUESTION: MCQ

Number of questions: 10 **Total mark: 8 X 1 + 1 X 2 = 10**

QUESTION 1: [1 mark]

State True or False.

The Continuous Bag of Words (CBOW) model predicts the surrounding context words given a center target word.

Correct Answer: False

Solution: CBOW uses a window of surrounding words to predict the target word whereas the Skip-gram model uses a target word to predict surrounding context words.

QUESTION 2: [1 mark]
What is the main drawback of representing words as one-hot vectors?
a. They cannot capture semantic similarity between words.
b. They are computationally inefficient.
c. They cannot incorporate word order effectively.
d. They are not robust to unseen words.

Correct Answer: a
Solution: One-hot vectors are orthogonal and do not capture relationships or similarities between words.

QUESTION 3: [1 mark]
A toy corpus has an initial base vocabulary of 6 unique characters: {a, b, c, g, s, t}. If the BPE algorithm performs exactly three merge operations, what will be the final size of the vocabulary?

a. 6
b. 3
c. 9
d. 10

Correct Answer: c
Solution: Starting with 6 characters and adding 3 new merged tokens results in a vocabulary size of 9.

QUESTION 4: [1 mark]
Which formula is used by WordPiece to compute the score of a pair of tokens (Token A and Token B) to determine if they should be merged?

a. Score = Frequency(A, B) + Frequency(A) + Frequency(B)
b. Score = Frequency(A, B) / (Frequency(A) x Frequency(B))

c. Score = Frequency(A, B) x (Frequency(A) + Frequency(B))
d. Score = (Frequency(A) x Frequency(B)) / Frequency(A, B)

Correct Answer: b

Solution: Please refer to lecture slides

QUESTION 5: [1 mark]

How does the WordPiece algorithm identify tokens during the tokenization of a new word?

a. It applies merge rules successively in the exact order they were learned during training.
b. It identifies the longest subword starting from the beginning of the word that exists in the vocabulary.
c. It uses the Viterbi algorithm to find the most probable sequence of tokens.
d. It randomly splits the word into subwords and calculates the frequency of each.
Correct Answer: b
Solution: Please refer to lecture slides.

QUESTION 6: [1 mark]
Which of the following statements are true about subword tokenization?

a. Frequently used words should be decomposed into smaller subwords to save space.
b. Rare words should be decomposed into meaningful subwords.
c. It serves as a middle ground between word-based and character-based tokenization.
d. It consists of two components: a token learner to generate a vocabulary and a token segmenter to process raw text.
Correct Answer: b, c, d
Solution: Please refer to lecture slides.

QUESTION 7: [2 marks]

For tf-idf weighting, if a word t appears in every single document in a collection of size N, what is its idf_t value?

a. 1
b. N
c. $\log_{10} N$
d. 0
Correct Answer: d
Solution: Please refer to lecture slides.

QUESTION 8: [1 mark]

What does the term "self-supervision" mean in the context of Word2Vec?
a. Using labels provided by human annotators
b. Automatically generating labels based on the relationship between words and their context
c. Supervising the training process manually
d. Employing external datasets for additional supervision

Correct Answer: b
Solution: Self-supervision in Word2Vec refers to the use of context words as labels to train the model without manual annotations.

QUESTION 9: [1 mark]
What is the key concept underlying Word2Vec?
a. Ontological semantics
b. Decompositional semantics
c. Distributional semantics
d. Morphological analysis

Correct Answer: c
Solution: Please refer to lecture slides.

Introduction to Large Language Models
Assignment- 5
TYPE OF QUESTION: MCQ

Number of questions: 8 **Total mark: 7 X 1 + 1 X 3 = 10**

QUESTION 1: [1 mark]
State True or False.
Increasing the temperature parameter during the decoding process makes the probability distribution sharper, resulting in a more confident and deterministic output.
a. True
b. False

Correct Answer: b
Solution: Please refer to lecture slides.

QUESTION 2: [3 marks]
Given the following encoder and decoder hidden states, compute the attention scores. (Use dot product as the scoring function)

Encoder hidden states: h_1=[2,1], h_2=[0,3], h_3=[4,2]

Decoder hidden state: s=[1,2]

a. 0.015, 0.843, 0.141
b. 0.843, 0.015, 0.141
c. 0.141, 0.843, 0.015
d. 0.016, 0.117, 0.867

Correct Answer: d
Solution:
$e_1=2 \cdot 1+1 \cdot 2=2+2=4$

$e_2=0 \cdot 1+3 \cdot 2=0+6=6$

$e_3=4 \cdot 1+2 \cdot 2=4+4=8$

$\alpha_1 = e^4/(e^4 + e^6 + e^8) = 0.0159$

$\alpha_2 = e^6/(e^4 + e^6 + e^8) = 0.1173$

$\alpha_3 = e^8/(e^4 + e^6 + e^8) = 0.867$

QUESTION 3: [1 mark]
What improvement does attention bring to the basic Seq2Seq model?
a. Reduces training time
b. Removes the need for an encoder
c. Allows access to all encoder states during decoding

d. Reduces the number of model parameters
Correct Answer: c

QUESTION 4: [1 mark]
Which gate in an LSTM is responsible for deciding how much of the cell state to keep?
a. Output gate
b. Input gate
c. Forget gate
d. Cell candidate gate
Correct Answer: c
Solution:
The forget gate determines what fraction of the previous cell state should be retained in the current timestep.

QUESTION 5: [1 mark]

What is the benefit of using multiple attention heads in a Transformer model?

a. It reduces the number of computations required.

b. It allows attending to multiple parts of the sequence from different representation subspaces.

c. It increases model sparsity.

d. It replaces the need for positional encodings.

Correct Answer: b

Solution: Each head can learn to focus on different types of relationships or dependencies within the input sequence.

QUESTION 6: [1 mark]

Why is masking used in the decoder part of a Transformer during training?

a. To prevent gradient explosion

b. To ensure the model does not attend to future tokens

c. To avoid overfitting

d. To normalize attention weights

Correct Answer: b

Solution: Causal masking ensures the decoder predicts one token at a time by masking out future positions.

QUESTION 7: [1 mark]

Why are RNNs preferred over fixed-window neural models?

a. They have a smaller parameter size.

b. They can process sequences of arbitrary length.

c. None of the above.

Correct Answer: b

Solution: Please refer to lecture slides.

QUESTION 8: [1 mark]

Which of the following is true about Seq2Seq models?
(i) Seq2Seq models are always conditioned on the source sentence.
(ii) The encoder compresses the input sequence into a fixed-size vector representation.
(iii) Seq2Seq models cannot handle variable-length sequences.
a. (i) and (ii)
b. (ii) only
c. (iii) only
d. (i), (ii), and (iii)

Correct Answer: a
Solution: Seq2Seq models are designed to encode variable-length sequences but compress them into fixed-size vector representations.

Introduction to Large Language Models
Assignment- 6
TYPE OF QUESTION: MCQ

Number of questions: 10 **Total mark: 10 X 1 = 10**

QUESTION 1: [1 mark]

State True or False:

In Transformers, masking is used only during inference to prevent looking at future tokens.

a. True

b. False

Correct Answer: b

Solution: Masking is used during both training and inference in autoregressive models to ensure that predictions depend only on past tokens.

QUESTION 2: [1 mark]

In self-attention, which of the following is true?

a. Queries and values are always equal

b. Each token attends only to the previous token

c. The output of each token is a weighted sum over all values

d. The key vectors are fixed after initialization

Correct Answer: c

QUESTION 3: [1 mark]

What is the purpose of masked decoding in Transformers?

a. Prevents attention lookups into the future

b. Discards irrelevant tokens from the sequence

c. Applies dropout to the attention weights

d. Limits the attention span of the model

Correct Answer: a

Solution: Please refer to the slides.

QUESTION 4: [1 mark]

For Rotary Position Embedding (RoPE), which of the following statements are true?

a. Combines relative and absolute positional information

b. Applies a multiplicative rotation matrix to encode positions

c. Eliminates the need for positional encodings

d. All of the above

Correct Answer: a, b

Solution: Please refer to the slides.

QUESTION 5: [1 mark]

Which of the following statements is NOT true?

a. Self-attention allows tokens to attend to every other token in the sequence.

b. Multi-head attention increases the overall dimensionality of the transformer model.

c. Relative positional encoding captures the distance between tokens rather than their absolute positions.

d. Positional encoding is necessary because self-attention alone is permutation invariant.

Correct Answer: b

Solution: Please refer to the slides.

QUESTION 6: [1 mark]

What is the primary motivation behind using transformers instead of RNNs or LSTMs?

a. To improve computational efficiency by avoiding sequential processing

b. To reduce the need for large datasets

c. To improve generalization to unseen data

d. To reduce the number of model parameters
Answer: a

QUESTION 7: [1 mark]

What is the primary reason for using layer normalization in transformers instead of batch normalization?

a. Layer normalization is faster than batch normalization.

b. Batch normalization is not effective for sequence models.

c. Layer normalization reduces overfitting.

d. Batch normalization does not work with positional encodings.
Answer: b

QUESTION 8: [1 mark]
Fill in the blank:
The sinusoidal positional encoding uses sine for even dimensions and ___ for odd dimensions.

a. sine
b. cosine
c. tangent
d. None of these

Answer: b

Solution: Please refer to lecture slides.

QUESTION 9: [1 mark]

You are given a self-attention layer with input dimension 2048, using 64 heads. What is the output dimension per head?

a. 64
b. 128
c. 32
d. 256

Answer: c
Solution: Each head processes 2048/64 = 32 dimensions

QUESTION 10: [1 mark]

_____ maps the values of a feature in the range [0,1].

a. Standardization
b. Normalization
c. Transformation
d. Scaling

Correct Answer: b

Solution: Normalization works by mapping all values of a feature to be in the range [0,1] using the transformation.

Introduction to Large Language Models
Assignment- 7

Number of questions: 9 **Total mark: 8 X 1 + 1 X 2 = 10**

QUESTION 1: [1 mark]

ELMo and BERT represent two different pre-training strategies for language models. Which of the following statement(s) about these approaches is/are true?

a) ELMo uses a bi-directional LSTM to pre-train word representations, while BERT uses a transformer encoder with masked language modeling.
b) ELMo provides context-independent word representations, whereas BERT provides context-dependent representations.
c) Pre-training of both ELMo and BERT involve next token prediction.
d) Both ELMo and BERT produce contextual embeddings usable for downstream tasks.

Correct Answer: a, d

Solution:

(a) **True.** ELMo's base model is a bidirectional language model implemented with stacked LSTMs (forward and backward LMs). BERT is a Transformer encoder trained with Masked Language Modeling (MLM) and Next Sentence Prediction (NSP).

(b) **False.** ELMo produces context-dependent token representations (its pooled representations depend on surrounding tokens via the BiLM). BERT also produces context-dependent representations. The statement claims ELMo is context-independent - that is incorrect.

(c) **False.** ELMo is trained with language-modeling objectives (predict next token in forward LM and previous token in backward LM), so it uses next-token prediction in its LM training. BERT, however, uses masked LM (MLM) rather than standard next-token prediction; it predicts a masked token given context. Thus "both involve next token prediction" is incorrect.

(d) **True.** Both produce contextual embeddings that downstream tasks can use and adapt: ELMo provides layerwise LM activations that tasks combine; BERT is fine-tuned, either end-to-end or an additional classification head is learnt, for tasks.

QUESTION 2: [1 mark]

Which model family is natively designed for conditional sequence-to-sequence tasks of the form P(y|x) without architectural modification?

a) Decoder-only language models
b) Encoder-only models
c) Encoder-decoder models
d) Character-CNN models

Correct Answer: c

Solution:

Encoder-decoder architectures (e.g., T5, BART) are built for seq2seq tasks: the encoder encodes x, the decoder autoregressively generates y conditioned on the encoder output. This matches P(y|x) natively. Decoder-only models can be prompted to approximate P(y|x) (with special concatenation/prefixing), but are not strictly architecturally seq2seq without prompts; encoder-only models (like BERT) do not generate sequences autoregressively by design.

QUESTION 3: [1 mark]

Which of the following corruption strategies was/were explored in the pre-training of BART?

a) Token masking, i.e., random tokens replaced by mask markers
b) Token deletion
c) Sentence permutation
d) Text infilling, i.e., replacing spans with a single sentinel and tasking decoder to generate the spans.

Correct Answer: a, b, c, d

Solution:

BART explored a family of noising/corruption strategies including token masking (random tokens replaced by mask markers), token deletion, sentence permutation (shuffling sentence order), and text infilling (replacing contiguous spans with sentinel tokens and asking decoder to reconstruct).

QUESTION 4: [1 mark]

What were the major changes introduced in GPT-2 family of models, compared to GPT-1?

a) Addition of layer normalization between sub-blocks
b) Increased vocabulary size
c) Increased context size
d) Addition of Rotary Positional Embedding (RoPE)

Correct Answer: a, b, c

Solution: GPT-2 kept the Transformer decoder architecture but introduced practical architectural and training changes: e.g., different layer-norm placement/tweaks (reported

differences in normalization), larger vocabulary (~50k WordPiece/GPT tokenizer variants), and larger context window (GPT-2 used up to 1024 tokens).

Rotary Positional Embeddings (RoPE) were however introduced later (not in GPT-2), so option (d) is false.

QUESTION 5: [1 mark]

Which of the following architectures are most suited for the sentiment classification task (given a sentence, assign one sentiment among positive, negative and neutral to it) and the machine translation task, respectively?

a) Sentiment Classification: Encoder-Decoder; Machine-Translation: Encoder-only
b) Sentiment Classification: Decoder-only; Machine-Translation: Encoder-only
c) Sentiment Classification: Encoder-only; Machine-Translation: Decoder-only
d) Sentiment Classification: Encoder-only; Machine-Translation: Encoder-Decoder

Correct Answer: d

Solution: For sentence-level classification (like sentiment classification), an encoder-only Transformer (e.g., BERT) is ideal: it creates a sentence representation used for a classification head.

For machine translation, a full encoder-decoder seq2seq model (e.g., BART/T5) is the natural match because it must map an input sequence in one language to an output sequence in another language. Thus, option (d) is correct.

QUESTION 6: [1 mark]

The chain-rule factorization used when training an autoregressive language model for a sentence $s = \{t_1, t_2, \dots, t_m\}$ leads to the log probability of s being expressed as:

a) $\log P(s) = \sum_{j=1}^{m} \log P(t_j)$
b) $\log P(s) = \log P(t_1) + \sum_{j=1}^{m-1} \log P(t_{j+1} | t_1, \dots, t_j)$
c) $\log P(s) = \log P(t_1) \times \prod_{j=1}^{m-1} \log P(t_{j+1} | t_1, \dots, t_j)$
d) $\log P(s) = \prod_{j=1}^{m} \log P(t_j)$

Correct Answer: b

Solution: Please refer to the lecture slides.

QUESTION 7: [1 mark]

Why does ELMo build its input token representations from a character-level CNN instead of fixed word embeddings?

a) To reduce training time by sharing parameters
b) To avoid UNK tokens and generate representations for any string
c) To compress embeddings to 128 dimensions
d) To ensure the same vector for a word in every context

Correct Answer: b

Solution: The main motivations for using a character-level CNN in ELMo are: (i) to avoid OOV/UNK tokens. Character composition lets the model form vectors for any token (including rare or unseen tokens), and (ii) to capture morphological patterns (prefixes, suffixes), reducing reliance on a fixed word vocabulary.

Options about compressing to 128 dims or guaranteeing same vector across contexts are incorrect (ELMo produces contextual vectors; the char-CNN simply forms a token representation before contextualization). So (b) is correct.

QUESTION 8: [1 mark]

In the ELMo task-specific combination formula, what is the role of the scalar γ^{task}?

a) It enforces a unit norm on the combined vector.
b) It selects which of forward/backward LSTMs to use.
c) It is an embedding for the task label.
d) It allows the downstream task to scale the overall magnitude of the combined representation.

Correct Answer: d

Solution: ELMo combines layer outputs as $ELMo_k^{task} = \gamma^{task} \sum_j s_j^{task} h_{k,j}^{LM}$. The learned scalar γ^{task} lets the downstream task adjust (scale) the overall magnitude of the combined representation. It is not a norm constraint or a selector of forward/backward LSTMs. So (d) is correct.

QUESTION 9: (Numerical Question) [2 marks]

The **einsum** function in numpy is used as a generalized operation for performing tensor multiplications. Now, consider two matrices: $A = \begin{bmatrix} 3 & 0 \\ 1 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & 8 \\ 7 & -7 \end{bmatrix}$. Then, what is the output of the following numpy operation?

```
numpy.einsum('ij,ji->', A, B)
```

Correct Answer: 9

Solution: `numpy.einsum('ij,ji->', A, B)` computes $\sum_i \sum_j A_{ij} B_{ji}$, i.e., the Frobenius inner product between A and B^T.

Thus the final answer is: [(3* (-2)) + (0*7) + (1*8) + ((-1)*(-7))] = -6 + 0 + 8 + 7 = 9.

Introduction to Large Language Models
Assignment- 8

Number of questions: 10 **Total mark: 10 X 1 = 10**

QUESTION 1: [1 mark]

Which factors influence the effectiveness of instruction tuning?

a) The number of instruction templates used in training.
b) The tokenization algorithm used by the model.
c) The diversity of tasks in the fine-tuning dataset.
d) The order in which tasks are presented during fine-tuning.

Correct Answers: a, c, d

Explanation:

1. **Number of instruction templates (a):** Having multiple, varied instruction templates helps the model learn how to follow instructions in different styles and formats. This increases the robustness of the model's instruction-following capability.

2. **Diversity of tasks in the fine-tuning dataset (c):** Exposing the model to a broad spectrum of tasks (e.g., classification, summarization, translation) teaches it to generalize better. If the dataset lacks diversity, the model risks overfitting to a narrow range of instructions.

3. **Order in which tasks are presented (d):** The sequence of tasks during multi-task or sequential fine-tuning can affect how well the model retains and integrates learned instructions. Certain curricula strategies (i.e., scheduling tasks in a deliberate sequence) often yield better performance.

4. **Why not the tokenization algorithm (b)?** While tokenization can impact model performance overall, it is not typically considered a primary factor in the effectiveness of instruction tuning itself. It plays a role in how text is broken down but is less critical than the factors in (a), (c), and (d).

QUESTION 2: [1 mark]

What are key challenges of soft prompts in prompt-based learning?

a) Forward pass with them is computationally inefficient compared to that with hard prompts.
b) They require additional training, unlike discrete prompts.
c) They cannot be interpreted or used effectively by non-expert users.
d) They require specialized architectures that differ from standard transformers.

Correct Answers: b, c

Explanation:

1. **They require additional training (b):** Soft prompts are continuous embeddings learned during training. Unlike hard (discrete) prompts, which are just tokens inserted into the text, soft prompts require gradient-based optimization. This necessitates a training procedure rather than a simple manual prompt.

2. **They cannot be interpreted or used effectively by non-expert users (c):** Since soft prompts are embedding vectors (continuous parameters), humans cannot directly read or interpret them. They are not as transparent or easily modifiable as typed-out text prompts.

3. **Why not (a) or (d)?**

   (a) Soft prompts do not necessarily result in a *computationally inefficient* forward pass compared to normal token embeddings. The computational overhead is minor and similar to standard embedding lookups.

   (d) Soft prompts generally *do not* require a specialized architecture; they can be applied to standard transformer-based models.

QUESTION 3: [1 mark]

Which statement best describes the impact of fine-tuning versus prompting in LLMs?

a) Fine-tuning is always superior to prompting in generalization tasks.
b) Prompting requires gradient updates, while fine-tuning does not.
c) Fine-tuning modifies the model weights permanently, while prompting does not.
d) Prompting performs better on in-domain tasks compared to fine-tuning.

Correct Answer: c

Explanation:

- **Fine-tuning modifies model weights permanently (c):** During fine-tuning, the model's parameters are updated, leading to a permanent change in its internal weights. In contrast, prompting relies on creating or inserting instructions (hard prompts) or learned embeddings (soft prompts) without altering the core model parameters.

- **Why not a, b, or d?**

  (a) Fine-tuning is not always superior to prompting; performance can vary based on task, data availability, and use case.

(b) Prompting usually does not require gradient updates (unless you are using soft prompts or another trainable method of prompting). Fine-tuning does require gradient updates.

(d) Prompting does not necessarily outperform fine-tuning on in-domain tasks; in some scenarios, a well-fine-tuned model can achieve higher in-domain accuracy.

QUESTION 4: [1 mark]

Which of the following aspects of the model outputs are captured by POSIX?

a) Diversity in the responses to intent-preserving prompt variations
b) Entropy of the distribution of response frequencies
c) Time required to generate responses for intent-preserving prompt variations
d) Variance in the log-likelihood of the same response for different input prompt variations

Correct Answer: a, b, d

Explanation:

1. **Diversity in responses (a):** POSIX aims to measure how many different valid responses the model produces under prompt variations that preserve the same intent.

2. **Entropy of response frequencies (b):** Entropy reflects how spread out or uncertain the model's responses are. POSIX takes into account how often each distinct response occurs, capturing response variability.

3. **Variance in log-likelihood under different prompts (d):** By measuring the log-likelihood of a particular response across varied prompts, POSIX can quantify how sensitive the model is to small changes in prompt phrasing.

4. **Why not (c)?**

   The time required for inference is not directly part of POSIX's metric. POSIX primarily focuses on response diversity and sensitivity to prompt variations, not computational speed.

QUESTION 5: [1 mark]

Which key mechanism makes Tree-of-Thought (ToT) prompting more effective than Chain-of-Thought (CoT)?

a) ToT uses reinforcement learning for better generalization.
b) ToT allows backtracking to explore multiple reasoning paths.
c) ToT reduces hallucination by using domain-specific heuristics.
d) ToT eliminates the need for manual prompt engineering.

Correct Answer: b

Explanation:

- **ToT allows backtracking (b):** Tree-of-Thought prompting structures the reasoning process in a tree-shaped manner, exploring multiple branches of thought. It can backtrack and test different solution paths, which helps uncover correct reasoning steps even if some initial paths were flawed.

- **Why not a, c, or d?**

  (a) While some implementations might use reinforcement learning, it is not the defining reason for ToT's effectiveness.

  (c) ToT does not inherently rely on domain-specific heuristics to reduce hallucination—rather, it systematically explores more possible reasoning paths.

  (d) ToT still benefits from careful prompt design for best results.

QUESTION 6: [1 mark]

What is a key limitation of measuring accuracy alone when evaluating LLMs?

a) Accuracy is always correlated with model size.
b) Accuracy cannot be measured on open-ended tasks.
c) Accuracy is independent of the training dataset size.
d) Accuracy does not account for prompt sensitivity.

Correct Answer: d

Explanation:

- **Accuracy does not account for prompt sensitivity (d):** Large Language Models can produce significantly different responses depending on how the prompt is phrased. Simply measuring accuracy on a set of prompts does not tell us whether the model's performance is stable across different prompt variations.

- **Why not a, b, or c?**

  (a) Accuracy is not always correlated with model size.

(b) Accuracy can be measured (in some form) even on open-ended tasks if appropriate metrics or reference answers are defined.

(c) Accuracy can be influenced by the training data size and quality. It is not truly independent of dataset size.

QUESTION 7: [1 mark]

Why is instruction tuning not sufficient for aligning large language models?

a) It does not generalize to unseen tasks.
b) It cannot prevent models from generating undesired responses.
c) It reduces model performance on downstream tasks.
d) It makes models less capable of learning from new data.

Correct Answer: b

Explanation:

- **It cannot prevent models from generating undesired responses (b):** While instruction tuning can help a model better follow user instructions, it does not fully address safe or ethical generation. A model might still produce harmful, biased, or factually incorrect outputs if not further aligned with techniques such as reinforcement learning from human feedback (RLHF) or safety guardrails.

- **Why not a, c, or d?**

  (a) Properly designed instruction tuning can help models generalize to new tasks, though not perfectly.

  (c) Instruction tuning does not inherently reduce performance on downstream tasks—often it aims to improve it.

  (d) Instruction tuning does not necessarily hinder a model's ability to learn from new data.

QUESTION 8: [1 mark]

Why is KL divergence minimized in regularized reward maximization?

a) To maximize the probability of generating high-reward responses.
b) To make training more computationally efficient.
c) To prevent the amplification of bias in training data.
d) To ensure models do not diverge too far from the reference model.

Correct Answer: d

Explanation:

- **Ensuring the model does not diverge too far (d):** Minimizing KL divergence between the updated policy (the newly optimized model) and a reference policy (the base model) constrains the model's changes. This helps the model exploit reward signals while retaining behaviours learned previously, avoiding catastrophic deviations that degrade performance or cause instability.

- **Why not a, b, or c?**

  (a) While reward maximization is the core goal, the reason to minimize KL is not specifically "to maximize high-reward responses" but to balance staying close to the original policy while moving toward higher reward.

  (b) KL divergence minimization does not necessarily make training more efficient computationally; it's more about controlling how drastically the policy updates.

  (c) Minimizing KL divergence can help with bias issues indirectly, but its primary purpose is to control the distribution shift rather than specifically address data bias.

QUESTION 9: [1 mark]

What is the primary advantage of using the log-derivative trick in REINFORCE?

a) Reducing data requirements
b) Expanding the token vocabulary
c) Simplifying gradient computation
d) Improving sampling diversity

Correct Answer: c

Explanation:

- **Simplifying gradient computation (c):** The log-derivative trick in the REINFORCE algorithm allows one to compute gradients of an expectation without having to differentiate the reward function directly (which might be discrete or non-differentiable). By bringing the log of the model's probability into the gradient expression, one can handle policy gradients in a more straightforward way.

- **Why not a, b, or d?**

  (a) The trick primarily addresses gradient calculation; it does not necessarily reduce how much data is needed.

  (b) It is unrelated to increasing the model's vocabulary.

(d) While it might affect how sampling is done during training, improving diversity is not its primary purpose.

QUESTION 10: [1 mark]

Which method combines reward maximization and minimizing KL divergence?

a) REINFORCE
b) Monte Carlo Approximation
c) Proximal Policy Optimization
d) Constitutional AI

Correct Answer: c

Explanation:

- **Proximal Policy Optimization (PPO) (c):** PPO is an algorithm designed to maximize a reward function while explicitly minimizing the KL divergence between the updated policy and the old policy. By doing so, it ensures that each update does not shift the policy too drastically, leading to more stable learning.

- **Why not a, b, or d?**

  (a) REINFORCE maximizes reward but does not inherently include a KL-based constraint to keep the new policy close to the old policy.

  (b) Monte Carlo Approximation is a technique, not a specific RL algorithm that enforces KL constraints.

  (d) Constitutional AI is a framework for aligning language models using a constitution of principles, not specifically about reward maximization combined with KL minimization in the same way PPO is.

Introduction to Large Language Models
Assignment- 9

Number of questions: 10 **Total mark: 10 X 1 = 10**

QUESTION 1: [1 mark]

Which of the following statement best describes why knowledge graphs (KGs) are considered more powerful than a traditional relational knowledge base (KB)?

a. KGs require no schema, whereas KBs must have strict schemas.
b. KGs store data only in the form of hypergraphs, eliminating redundancy.
c. KGs allow flexible, graph-based connections and typed edges, enabling richer relationships and inferences compared to KBs.
d. KGs completely replace the need for textual sources by storing all possible facts.

Correct Answer: c

Explanation:

- Traditional relational knowledge bases enforce strict schemas (rows, columns, tables). In contrast, KGs store entities as nodes with typed edges (relations) between them, allowing richer and more flexible relationships.

- KGs can represent complex connections (e.g., multi-edges, different relation types) and support inference by traversing these connections.

- While some knowledge graphs can be partially schema-less or schema-flexible, choice (c) specifically highlights the flexibility in graph connections and typed edges, which is what makes KGs generally more powerful for representing rich, interlinked data.

QUESTION 2: [1 mark]

Entity alignment and relation alignment are crucial between KGs of different languages. Which of the following factors contribute to effective alignment?

a. Aligning relations solely by their lexical similarity, ignoring semantic context
b. Transliteration or language-based string matching for entity labels
c. Ensuring all language aliases are represented identically in each KG
d. Matching neighbours, or connected entities, across different KGs

Correct Answer: b, d

Explanation:

- **Transliteration or language-based string matching (b):** For multilingual KGs, matching entity labels across languages (e.g., transliterating names or matching synonyms) is key to identifying equivalent entities.

- **Matching neighbours (d):** Alignment goes beyond just matching labels; you also look at the graph structure - if two entities are neighbours or connected to the same concept in each KG, that supports they might be the same or related.

- **Why not a or c?**

  (a) Lexical similarity alone (ignoring context) is insufficient, as terms might match lexically but differ in meaning across languages.

  (c) It is often not possible or necessary to have identical labels for all language aliases if they mean the same thing in context; the important aspect is linking them, not forcing identical representations.

QUESTION 3: [1 mark]

In the context of knowledge graph completion (KGC), which statement best describes the role of the scoring function $f(s, r, o)$?

a. It determines whether two entities refer to the same real-world concept.
b. It produces a raw confidence score indicating how plausible a triple (s, r, o) is.
c. It explicitly encodes only the subject's embedding, ignoring the relation and object embeddings.
d. It ensures that every negative triple gets a higher score than any positive triple.

Correct Answer: b

Explanation:

- **A raw confidence score for plausibility (b):** In KGC, the scoring function evaluates a triple (s, r, o) (subject, relation, object) to determine how likely it is to be true according to the learned embeddings. A higher (or lower, depending on the convention) score indicates higher plausibility.

- **Why not the others?**

  (a) That relates to entity resolution, not necessarily the KGC scoring function.

  (c) The scoring function typically factors in all three embeddings (subject, relation, and object).

(d) While in training we often prefer valid triples to have higher scores than negative ones, the scoring function itself just produces a plausibility value. It doesn't guarantee all negative triples score higher or lower unconditionally.

QUESTION 4: [1 mark]

One key difference between the differentiable KG approach and the semantic interpretation approach to KGQA is:

a. Differentiable KG approaches are fully rule-based, while semantic interpretation is purely neural.
b. Differentiable KG approaches do not require any graph embeddings, relying instead on explicit logical forms.
c. Semantic interpretation is more transparent or interpretable, whereas differentiable KG is end-to-end trainable but less interpretable.
d. Both approaches use logical forms; the primary difference is the type of question they can answer.

Correct Answer: c

Explanation:

- **Semantic interpretation is more interpretable, differentiable KG is more end-to-end (c):**
    - Semantic interpretation methods typically rely on building an explicit logical form of the question, which is transparent and can be easily explained.
    - Differentiable KGQA uses neural embeddings and end-to-end backpropagation over the graph, making it powerful but less human-interpretable.

- **Why not the others?**

  (a) Differentiable KG approaches are *not* fully rule-based — in fact, they're more neural and less rule-based.

  (b) Differentiable approaches definitely *do* use graph embeddings.

  (d) While both can handle complex questions, the key difference in c is the interpretability vs. end-to-end trainability.

QUESTION 5: [1 mark]

Considering the differentiable KG approach, which elements are typically learned jointly when training an end-to-end KGQA model?

a. The textual question representation (e.g., BERT embeddings)
b. The graph structure encoding (e.g., GCN or transformer-based graph embeddings)
c. Predefined logical forms to ensure interpretability
d. The final answer selection mechanism that identifies which node(s) in the graph satisfy the question

Correct Answer: a, b, d

Explanation:

- **Textual question representation (a):** The system learns how to embed the input question (often with a neural model like BERT).
- **Graph structure encoding (b):** It also learns how to encode nodes and relations in the knowledge graph (using a graph neural network, attention, etc.).
- **Final answer selection (d):** Finally, the model learns how to map from the question and graph embeddings to the correct node(s) in the KG.
- **Why not (c)?** Predefined logical forms are more typical of semantic-parsing-based KGQA, not differentiable KGQA approaches. Differentiable KGQA is usually end-to-end and does not require manually crafted logical forms.

QUESTION 6: [1 mark]

Uniform negative sampling can have high variance and may require large number of samples. Why is that the case?

a. Because the margin-based loss cannot converge without big mini-batches.
b. Because randomly picking negative entities does not guarantee close or challenging negatives, causing unstable training estimates.
c. Because negative sampling must ensure every possible negative triple is covered.
d. Because the number of relations in the KG is too large for small number of samples.

Correct Answer: b

Explanation:

- **High variance arises when negatives are not challenging (b):** If negative examples are chosen completely at random, many will be too easy for the model to distinguish, providing limited learning signal. The model sees less “borderline” cases, so estimates of how well the model can separate real vs. fake facts fluctuate significantly.

- **Why not the others?**

  (a) Margin-based losses can converge with or without large mini-batches if sampling is done carefully.

(c) We don't need to cover every possible negative triple, just enough meaningful ones for training.

(d) The number of relations in a KG might be large, but that alone doesn't necessarily drive the variance issue.

QUESTION 7: [1 mark]

In testing embedding and score quality for KG completion, *mean rank* and *hits@K* are typical metrics. What does *hits@K* specifically measure in this context?

a. The percentage of queries for which the correct answer appears in the top-K of the ranked list.
b. The reciprocal of the rank of the correct answer.
c. The probability of the correct answer appearing as the highest scored candidate.
d. The margin of the correct triple score relative to all negative triples.

Correct Answer: a

Explanation:

- **Hits@K = The percentage of queries for which the correct entity (or triple) is in the top-K predictions (a)**. This means if the correct answer is within the first K results in the ranking, we call it a "hit." We then compute how many queries achieve this, divided by the total.

- **Why not b, c, or d?**

  (b) That is more like Mean Reciprocal Rank (MRR), not hits@K.

  (c) If K=1, that might coincide with hits@1, but the metric hits@K is about the top-K in general, not exclusively the top candidate.

  (d) That describes a margin-based idea, not hits@K.

QUESTION 8: [1 mark]

In the TransE model, the scoring function for a triple (s, r, o) is typically defined as

$$f(s, r, o) = \| e_s + e_r - e_o \|$$

where e_s, e_r, e_o are embeddings of the subject, relation, and object, respectively. Which statement best explains what a low value of $f(s, r, o)$ indicates in this context?

a. That (s, r, o) is an invalid triple according to the learned embeddings.

b. That e_s and e_o must be orthogonal.
c. That the relation embedding e_r is zero.
d. That (s, r, o) has a high likelihood of being a true fact in the knowledge graph.

Correct Answer: d

Explanation:

- **A low distance = a high plausibility (d):** In TransE, the model is trained such that $s + r \approx o$ for a valid triple. If the norm $\| s + r - o \|$ is small, it means the subject, relation, and object embeddings line up well, indicating that triple is likely true.

- **Why not a, b, c?**

  (a) A high value would correspond to an invalid triple.

  (b) Orthogonality is not directly indicated by a small distance in TransE.

  (c) Zero relation embedding is not required for plausibility; the relation embedding can be non-zero and still yield a low distance.

QUESTION 9: [1 mark]

In RotatE, if a relation r is intended to be symmetric, how would that typically manifest in the complex plane?

a. The relation embedding e_r must always equal zero.
b. The angle of e_r must be $\frac{\pi}{2}$.
c. The relation embedding e_r is its own inverse (i.e., a 180° rotation when squared).
d. The magnitude of e_r must be greater than 1.

Correct Answer: c

Explanation:

- **Relation embedding is its own inverse (c):** In RotatE, each relation is modeled as a rotation in the complex plane. For a relation to be symmetric, applying that relation twice would yield the original entity, so $r^2 = 1$. A 180° rotation (i.e., π radians) is its own inverse because rotating twice by 180° brings you back to the same orientation.

- **Why not a, b, or d?**

  (a) A zero embedding is not characteristic of symmetry in RotatE.

  (b) An angle of π radians alone indicates an inverse relation, but the question specifically references the notion of the embedding being its own inverse, which implies $r^2 = 1$.

(d) The magnitude constraint (often magnitude = 1 in RotatE) is not specifically about symmetry.

QUESTION 10: [1 mark]

Which main advantage do rotation-based models (like RotatE) have over translation-based ones (like TransE) when it comes to complex multi-relational patterns in a KG?

a. Rotation-based models cannot model any symmetry or inverse patterns, so they are simpler.
b. Rotation-based models handle a broader set of relation properties (symmetry, anti-symmetry, inverses, composition) more naturally.
c. Rotation-based models have no hyperparameters to tune, unlike TransE.
d. Rotation-based models are guaranteed to yield perfect link prediction.

Correct Answer: b

Explanation:

- **Rotation-based models can capture more complex relational properties (b):** By representing relations as rotations in the complex plane, RotatE naturally supports symmetric relations ($r^2 = 1$) anti-symmetric relations, inverses (rotations in the opposite direction), and composition (cumulative rotations).

- **Why not a, c, or d?**

  (a) In fact, they can model symmetry, inverses, etc.

  (c) They do have hyperparameters (e.g., embedding dimension, learning rate); it is not hyperparameter-free.

  (d) No model is guaranteed to be perfect for all link prediction tasks.

Introduction to Large Language Models
Assignment- 10

Number of questions: 10 **Total mark: 10 X 1 = 10**

QUESTION 1: [1 mark]

How do Prefix Tuning and Adapters differ in terms of where they inject new task-specific parameters in the Transformer architecture?

a. Prefix Tuning adds new feed-forward networks after every attention block, while Adapters prepend tokens.
b. Both approaches modify only the final output layer but in different ways.
c. Prefix Tuning learns trainable “prefix” hidden states at each layer’s input, whereas Adapters insert small bottleneck modules inside the Transformer blocks.
d. Both approaches rely entirely on attention masks to inject new task-specific knowledge.

Correct Answer: c

Explanation:

- **Prefix Tuning:** In this approach, we learn a sequence of “prefix” embeddings - trainable hidden states - that get prepended to the model’s internal representations at each layer. This means the main Transformer weights stay frozen, and the prefix acts like extra context or “virtual tokens” that the model attends to.

- **Adapters:** Adapters insert small, trainable modules (often feed-forward “bottleneck” layers) inside each Transformer layer, typically after the attention or feed-forward sub-layers. They are small enough to keep the main model weights mostly intact while still enabling fine-tuning for new tasks.

Therefore, the main difference is that Prefix Tuning adds trainable “prefix” embeddings at the input side of each layer, while Adapters add small modules inside the architecture.

QUESTION 2: [1 mark]

The Structure-Aware Intrinsic Dimension (SAID) improves over earlier low-rank adaptation approaches by:

a. Ignoring the network structure entirely
b. Learning one scalar per layer for layer-wise scaling
c. Sharing the same random matrix across all layers
d. Using adapters within self-attention layers

Correct Answer: b

Explanation:

- **Structure-Aware Intrinsic Dimension (SAID):** This method learns a small set of parameters - often including one scalar per layer - to scale or adjust each layer. Instead of ignoring the model's structure, it captures the global scaling behaviour across layers with minimal additional parameters.

- **Why not the others?**

  (a) SAID explicitly does not ignore the network structure; it leverages per-layer learnable parameters.

  (c) Sharing the exact same random matrix across all layers is a different approach (not specifically SAID's main method).

  (d) SAID is more about scalar scaling factors, not inserting adapter modules.

QUESTION 3: [1 mark]

Which of the following are correct about the extensions of LoRA?

a. LongLoRA supports inference on longer sequences using global attention
b. QLoRA supports low-rank adaptation on 4-bit quantized models
c. DyLoRA automatically selects the optimal rank during training
d. LoRA+ introduces gradient clipping to stabilize training

Correct Answer: b, c

Explanation:

- **QLoRA (b)** applies the LoRA idea to a quantized model, typically at 4-bit precision, enabling efficient fine-tuning with significantly reduced memory use.
- **DyLoRA (c)** is an approach that dynamically picks the optimal rank during training, thereby adapting the low-rank decomposition to the task at hand.

- **Why not (a) or (d)?**

  (a) "LongLoRA" is not a commonly recognized extension that specifically implements global attention for longer sequences as part of LoRA. (Some methods support longer-context inference, but that is not typically referred to as "LongLoRA.")

  (d) While "LoRA+" might be used informally in some contexts, the "+" here does not refer to a widely acknowledged official extension that only introduces gradient clipping.

QUESTION 4: [1 mark]

Which pruning technique specifically removes weights with the smallest absolute values first, potentially followed by retraining to recover accuracy?

a. Magnitude Pruning
b. Structured Pruning
c. Random Pruning
d. Knowledge Distillation

Correct Answer: a

Explanation:

- **Magnitude Pruning** removes weights whose absolute values are below a certain threshold (i.e., the smallest magnitudes). The rationale is that weights with small magnitudes contribute less to overall model outputs. After pruning them, one can optionally retrain (also called "fine-tuning" after pruning) to recover lost accuracy.

- **Structured Pruning (b)** removes entire groups of weights (e.g., entire filters or channels).

- **Random Pruning (c)** removes weights randomly, with no magnitude criterion.

- **Knowledge Distillation (d)** is a different approach, transferring knowledge from a teacher to a student model, not a direct pruning method.

QUESTION 5: [1 mark]

In Post-Training Quantization (PTQ) for LLMs, why is a calibration dataset used?

a. To precompute the entire attention matrix for all tokens.
b. To remove outlier dimensions before applying magnitude-based pruning.
c. To fine-tune the entire model on a small dataset and store the new weights.
d. To estimate scale factors for quantizing weights and activations under representative data conditions.

Correct Answer: d

Explanation:

- **Calibration Dataset:** In PTQ, you typically don't retrain the model. Instead, you gather a small "calibration set" of representative examples. By running these examples through the model, you observe the distribution of activations (and possibly weights). This helps you pick appropriate *scale* factors (or quantization parameters) so that the quantized model preserves accuracy as much as possible.

- Thus, it's about **extracting distribution statistics** to set the quantization scale and zero points.

QUESTION 6: [1 mark]

Which best summarizes the function of the unembedding matrix W_U?

a. It merges the queries and keys for each token before final classification.
b. It converts the final residual vector into vocabulary logits for next-token prediction.
c. It is used for normalizing the QK and OV circuits so that their norms match.
d. It acts as a second attention layer that aggregates multiple heads.

Correct Answer: b

Explanation:

- The **unembedding matrix** (often the transpose of the embedding matrix – if weight sharing is enabled) takes the final hidden state (i.e., the final residual or contextual representation of each token) and maps it to a distribution over the vocabulary. That distribution is used to pick the next token.

- **Why not the others?**

  (a) Merging queries and keys is part of the attention mechanism, not the unembedding step.

  (c) Normalizing QK or OV circuits usually involves layer norms or other scaling parameters, not the unembedding.

  (d) A second attention layer is not typically called the "unembedding" layer.

QUESTION 7: [1 mark]

Which definition best matches an induction head as discovered in certain Transformer circuits?

a. A head that specifically attends to punctuation tokens to determine sentence boundaries
b. A feed-forward sub-layer specialized for outputting next-token probabilities for out-of-distribution tokens
c. A head that looks for previous occurrences of a token A, retrieves the token B that followed it last time, and then predicts B again
d. A masking head that prevents the model from looking ahead at future tokens

Correct Answer: c

Explanation:

- **Induction heads**: These specialized attention heads implement a pattern that can be described like: “If we see a token A repeated, we look back to see what token came after A the last time it appeared, and we hypothesize that token will appear again.” This mechanism is behind repeating patterns or reusing local context the model has seen before.
- It is effectively a memory pattern that picks up repeated sequences in text.

QUESTION 8: [1 mark]

In mechanistic interpretability, how can we define ‘circuit’?

a. A data pipeline for collecting training examples in an autoregressive model
b. A small LSTM module inserted into a Transformer for additional memory
c. A device external to the neural network used to fine-tune certain parameters after training
d. A subgraph of the neural network hypothesized to implement a specific function or behaviour

Correct Answer: d

Explanation:

- In **mechanistic interpretability**, a *circuit* is a **subgrap**h of a neural network—specific connections and components (e.g., heads, neurons, MLP layers)—that collectively implement a certain interpretable function. Researchers attempt to identify and visualize these circuits to understand *how* the model handles specific patterns or tasks.

QUESTION 9: [1 mark]

Which best describes the role of Double Quantization in QLoRA?

a. It quantizes the attention weights twice to achieve 1-bit representations.
b. It reinitializes parts of the model with random bit patterns for improved regularization.
c. It quantizes the quantization constants themselves for additional memory savings.
d. It systematically reverts partial quantized weights back to FP16 whenever performance degrades.

Correct Answer: c

Explanation:

- **Double Quantization** in **QLoRA** means not only are the main weights quantized, but the scaling factors (quantization constants) are themselves stored in a lower precision format to reduce memory usage further. It's essentially a second layer of quantization on top of the initial quantization scheme, yielding additional memory compression.

- **Why not the others?** There's no step of reinitializing with random bit patterns, nor switching to 1-bit, nor automatically reverting to FP16.

QUESTION 10: [1 mark]

Which of the following are true about sequence-level distillation for LLMs?

a. It trains a student model by matching the teacher's sequence outputs (e.g., predicted token sequences) rather than just individual token distributions.
b. It requires storing only the top-1 predictions from the teacher model for each token.
c. It can be combined with word-level distillation to transfer both local and global knowledge.
d. It forces the teacher to produce a chain-of-thought explanation for each example.

Correct Answer: a, c

Explanation:

- **Sequence-level distillation:** Instead of just matching the teacher's probability distribution at each token, the student is trained to mimic the teacher's entire output sequence (which can capture *global*, multi-token patterns).

    - **(a) True**: The student tries to match the teacher's full sequence outputs, not just per-token probabilities in isolation.

    - **(c) True:** Combining sequence-level with word-level distillation can yield a comprehensive approach where the student learns local token distribution and overall sequence structure.

- **Why not (b) or (d)?**

    - (b) Typically, sequence-level distillation can store more than top-1 predictions (e.g., entire sequences). Just top-1 might lose richer distribution information.

    - (d) It does *not* necessarily require chain-of-thought. Sequence-level distillation focuses on final outputs; it doesn't force an intermediate chain-of-thought explanation.

Introduction to Large Language Models
Assignment- 11

Number of questions: 10 **Total mark: 10 X 1 = 10**

QUESTION 1: [1 mark]

What is the main modification that SimplE makes to DistMult-like models to handle asymmetric relations?

a. Replacing entity embeddings with random fixed vectors
b. Introducing separate entity embeddings for subject and object roles, along with inverse relations
c. Restricting the rank of the relation tensor to 1
d. Using negative sampling for half of the triple set

Correct Answer: b

Explanation:

- **SimplE** extends the DistMult approach by using **two different embeddings** for each entity: one as a subject (head) and one as an object (tail). It also learns inverse relation embeddings to handle asymmetry.

- DistMult alone struggles with asymmetric relations because its scoring function $score(s, r, o)$ is symmetrical in s and o. SimplE partially addresses this by modelling subject/object roles distinctly and considering inverse relations.

QUESTION 2: [1 mark]

Which statements correctly characterize the basic DistMult approach for knowledge graph completion?

a. Each relation r is parameterized by a full D×D matrix that can capture asymmetric relations.
b. The relation embedding is a diagonal matrix, leading to a multiplicative interaction of entity embeddings.
c. DistMult struggles with non-symmetric relations because score(s, r, o) = $a_s^T M_r a_o$ is inherently symmetric in s and o.
d. DistMult's performance is typically tested only on fully symmetric KGs.

Correct Answer: b, c

Explanation:

- **DistMult** stands for "*Diagonal* bilinear model", meaning each relation r has a diagonal embedding M_r. This is effectively a vector multiplied element-wise (since a diagonal matrix just scales each coordinate).

- This diagonal constraint leads to symmetric scoring: swapping s and o yields the same value because of the multiplication pattern.

- Therefore:

    - (b) is **true**: the relation embedding is diagonal.

    - (c) is **true**: it struggles with asymmetric relations because $a_s^T(diag(r))a_o$ is symmetric in a_s and a_o.

- (a) is **false** because DistMult does not use a full D×D matrix; it specifically uses a diagonal matrix.

- (d) is also **false**: DistMult can be tested on general KGs (not only fully symmetric ones), though it does poorly with strongly asymmetric relations.

QUESTION 3: [1 mark]

Which statements about the ComplEx extension of DistMult are true?

a. It uses complex-valued embeddings to better capture asymmetric or anti-symmetric relations.
b. It replaces the multiplication in DistMult with element-wise addition of real-valued vectors.
c. For a perfectly symmetric relation, one could set the imaginary part of the relation embedding to zero.
d. ComplEx requires each entity vector to be unit norm in the complex plane.

Correct Answer: a, c

Explanation:

ComplEx extends DistMult to the complex domain, allowing it to handle asymmetric relations by using conjugate-based interactions. A typical ComplEx scoring function is: $Re(< a_s,\ r,\ \overline{a_o} >)$, where a_s, r, a_o are complex vectors, and $\overline{a_o}$ is the complex conjugate of a_o .

- (a) **True**: The use of complex embeddings allows capturing asymmetry via imaginary components and conjugation.
- (c) **True:** If the imaginary part is zero, the relation effectively behaves like DistMult in the real domain, which is symmetric.

- (b) is **false**: It still uses multiplicative interactions, not element-wise addition.

- (d) is **not necessarily required**; while some implementations might normalize embeddings, ComplEx does not strictly require each entity to have unit norm. It's a common training choice but not an inherent necessity.

QUESTION 4: [1 mark]

Which best describes the main advantage of using a factorized representation (e.g., DistMult, ComplEx) for large KGs?

a. It enforces that every relation in the KG be perfectly symmetric.
b. It ensures each entity is stored as a one-hot vector, simplifying nearest-neighbour queries.
c. It collapses the entire KG into a single scalar value.
d. It significantly reduces parameters and enables generalization to unseen triples by capturing low-rank structure.

Correct Answer: d

Explanation:

- Factorized models (DistMult, ComplEx, etc.) leverage low-rank embeddings for entities and relations, drastically reducing the parameters needed compared to storing a full adjacency or full 3D tensor for the entire knowledge graph.

- This **low-rank structure** also helps the model learn patterns and generalize to unseen triples (i.e., it can guess new edges that weren't explicitly in training).

QUESTION 5: [1 mark]

Which statement best describes the *reshaping* of a 3D KG tensor $X \in R^{|E|\times|R|\times|E|}$ into a matrix factorization problem?

a. One axis remains for subject, one axis remains for object, and relations are combined into a single expanded axis.
b. The subject dimension is repeated to match the relation dimension, resulting in a 2D matrix.
c. Each subject–relation pair is collapsed into a single dimension, while objects remain as separate entries.
d. The entire KG is vectorized into a 1D array and then factorized with an SVD approach.

Correct Answer: c

Explanation:

- A 3D KG tensor $X \in R^{|E|\times|R|\times|E|}$ typically has subject-relation-object axes.

- One common approach is to "unfold" or "reshape" the tensor along one axis to produce a matrix. E.g., you can merge the subject and relation dimensions into a single dimension (subject-relation pairs) and keep the object axis separate, yielding a 2D matrix factorization problem.

- This approach is used in some methods to apply standard matrix factorization to the KG data.

QUESTION 6: [1 mark]

Which key property of hierarchical relationships (e.g. is-a, transitivity) motivates the exploration of specialized embedding methods over standard Euclidean KG embeddings?

a. Symmetry in the relation (A, is-a, B) implying (B, is-a, A)
b. Frequent presence of cycles in hierarchical graphs
c. Transitivity in the form (camel, is-a, mammal) and (mammal, is-a, animal) $\Longrightarrow$ (camel,is-a,animal)
d. The high dimensionality of the entity embeddings

Correct Answer: c

Explanation:

- **Hierarchical** "is-a" relations typically exhibit **transitivity**: if A is-a B, and B is-a C, then A is-a C. Standard Euclidean embeddings (like DistMult, ComplEx) do not directly enforce or encourage transitivity.
- Specialized approaches (e.g., hyperbolic embeddings, order embeddings) do a better job of **naturally capturing** such transitive relations.

QUESTION 7: [1 mark]

Which of the following statements correctly describe hyperbolic (Poincare) embeddings for hierarchical data?

a. They map nodes onto a disk (or ball) such that large branching factors can be represented with lower distortion than in Euclidean space.
b. Distance grows slowly near the centre and becomes infinite near the boundary, making it naturally suited for tree-like structures.
c. They require each node to be embedded on the surface of the Poincare disk of radius 1.
d. They can achieve arbitrarily low distortion embeddings for trees with the same dimension as Euclidean space.

Correct Answers: a, b

Explanation:

- **Hyperbolic (Poincaré) embeddings** place points in a disk (2D) or ball (higher-D) where distances near the boundary can become very large, effectively allowing the model to represent tree-like expansions with less distortion than Euclidean.

- Specifically:

    - (a) **True:** Large branching structures are more compactly represented.

    - (b) **True:** Distances blow up near the boundary, which helps model hierarchical "depth."

- (c) is **not** strictly correct: points reside within the disk, not necessarily on the boundary (though some parameterizations might keep norms < 1).

- (d) is also not entirely correct in claiming "arbitrarily low distortion in the *same* dimension." Hyperbolic embeddings do better than Euclidean in capturing tree metrics, but **not** everything is **guaranteed** "arbitrarily low" in the same dimension. They do reduce distortion significantly compared to Euclidean though.

QUESTION 8: [1 mark]

Why might a partial-order-based approach (like order embeddings) be beneficial for modelling 'is-a' relationships compared to purely distance-based approaches?

a. They explicitly encode the ancestor–descendant relation as a coordinate-wise inequality or containment.
b. They can represent negative correlations (i.e., sibling vs. ancestor) more easily than distance metrics.
c. They inherently guarantee transitive closure of the hierarchy in the learned embedding space.
d. They do not rely on pairwise distances but use a notion of coordinate-wise ordering or interval containment.

Correct Answer: a, d

Explanation:

- **Order embeddings** interpret the "is-a" relationship as a partial order, typically using **coordinate-wise constraints** (for instance, $X \leq Y$ if $x_i \leq y_i \ \forall i$). This can capture ancestor-descendant with a straightforward geometric interpretation (e.g., intervals, boxes, or cones).
- They do not rely solely on a distance measure; they rely on the geometry of "containment" or inequality in each coordinate.

- Hence, (a) and (d) are correct.

- (b) is not necessarily guaranteed or specifically a reason they do it more easily; negative correlation is a different concept.

- (c) They do not *automatically* guarantee full transitive closure in the sense that it's not a built-in "guarantee." The embeddings typically encourage partial ordering, but some additional constraints or training objectives are needed.

QUESTION 9: [1 mark]

Which statement about box embeddings in hierarchical modelling is most accurate?

a. Each entity or type is assigned a single real-valued vector, ignoring bounding volumes.
b. Containment $I_x \subseteq I_y$ all dimensions encodes $x \prec y$.
c. They rely on spherical distances around a central node to measure tree depth.
d. They cannot be used to represent set intersections or partial overlap.

Correct Answer: b

Explanation:

- **Box Embeddings** assign each concept or entity a "box" in some coordinate space, typically described by two vectors $l \leq u$ (lower and upper corners).

- If box X is contained within box Y on every coordinate dimension (i.e., $l_X \geq l_Y$ and $u_X \leq u_Y$), that corresponds to an "is-a" or subset relationship.

- This approach can also represent partial overlaps (intersecting boxes) and so on, so (d) is incorrect.

- (c) describes a more spherical or Euclidean method, not boxes.

- (a) is the opposite of the box approach (which does *not* ignore bounding volumes but explicitly uses them).

QUESTION 10: [1 mark]

What is a key challenge with axis-aligned open-cone (order) embeddings for hierarchical KG data?

a. They enforce that all sibling categories have identical cone apices, which causes overlap.
b. They require symmetrical relationships for all edges.

c. They do not allow partial orders to be extended to total orders.
d. The volume (measure) of cones is the same regardless of how “broad” or “narrow” the cone is, making sub-categories indistinguishable by volume.

Correct Answer: d

Explanation:

- In **axis-aligned cone embeddings**, each entity is associated with an open cone in a coordinate space. The cones can be wide or narrow, but the measure of the open cone does not reflect “how big” or “small” the category is, which makes it difficult to distinguish sub-categories based on “volume.”

- This is a known limitation: the geometry does not allow easily using volume-based notions to reflect hierarchical or subset relationships beyond the angular direction.